THE FLAT PULLBACK COMMUTES WITH THE BASE CHANGE PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FVK (sources/stacks-chow/pinned-original-source.tex line 15302).

Allowed standard prerequisites (editorial).

- The base change pullback on cycles and Chow groups with its degree shift. Same chapter, Lemma labelled lemma-pullback-base-change, invoked in the statement, and the construction of g^* at proof lines 15329-15331. Supplies the horizontal maps of the diagrams.
- The flat pullback on cycles. Same chapter, Definition labelled definition-flat-pullback, cited at proof lines 15333-15335. Gives f^*[Z] = [Z x_X Y]_{k+r}.
- Base change of the coherent pullback of a cycle. Same chapter, Lemmas labelled lemma-pullback-coherent and lemma-pullback-coherent-base-change, cited at proof lines 15331-15337. Computes the two composites of the cycle level square.

Source reference. Original reference 0FVG: sources/stacks-chow/pinned-original-source.tex, source lines 15123–15123. Same-chapter pointer into the author environment situation at source line 15123; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02RE: sources/stacks-chow/pinned-original-source.tex, source lines 2331–2331. Same-chapter pointer into the author environment lemma at source line 2331; invoked at proof line 15329 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 02RB: sources/stacks-chow/pinned-original-source.tex, source lines 2219–2219. Same-chapter pointer into the author environment definition at source line 2219; invoked at proof line 15332 of the counted

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proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FVI: sources/stacks-chow/pinned-original-source.tex, source lines 15193–15193. Same-chapter pointer into the author environment lemma at source line 15193; invoked at proof line 15333 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0FVG let $Y \to X \to S$ be locally of finite type and let $Y' \to X' \to S'$ be the base change by $S' \to S$. Assume $f : Y \to X$ is flat of relative dimension r. Then $f' : Y' \to X'$ is flat of relative dimension r and the diagrams*

$$\begin{array}{ccc} Z_{k+r}(Y) & \xrightarrow[g^*]{} & Z_{k+c+r}(Y') \\ {\scriptstyle f^*}\uparrow & & \uparrow{\scriptstyle (f')^*} \\ Z_k(X) & \xrightarrow{g^*} & Z_{k+c}(X') \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathrm{CH}_{k+r}(Y) & \xrightarrow[g^*]{} & \mathrm{CH}_{k+c+r}(Y') \\ {\scriptstyle f^*}\uparrow & & \uparrow{\scriptstyle (f')^*} \\ \mathrm{CH}_k(X) & \xrightarrow{g^*} & \mathrm{CH}_{k+c}(X') \end{array}$$

of cycle and chow groups commutes.

Proof. It suffices to show the first diagram commutes. To see this, let $Z \subset X$ be an integral closed subscheme of δ-dimension k and denote $Z' \subset X'$ its base change. By construction we have $g^*[Z] = [Z']_{k+c}$. By Lemma 02RE we have $(f')^* g^*[Z] = [Z' \times_{X'} Y']_{k+c+r}$. Conversely, we have $f^*[Z] = [Z \times_X Y]_{k+r}$ by Definition 02RB. By Lemma 0FVI we have $g^* f^*[Z] = [(Z \times_X Y)']_{k+r+c}$. Since $(Z \times_X Y)' = Z' \times_{X'} Y'$ by associativity of fibre product we conclude. □